

3. Methodology

This report uses a quantitative, comparative, secondary-data approach to analyse progress toward Universal Health Coverage under SDG 3.8. The aim is to compare progress across countries, regions, and income groups using internationally comparable indicators, rather than to evaluate one national case in depth (United Nations, 2015; World Health Organization and World Bank, 2025). Secondary data were therefore appropriate because they allowed broad country coverage, long-term comparison, and alignment with the approved international data sources highlighted in the project brief and kickoff guidance.

The analysis is based on three core indicators that reflect the main dimensions of UHC. The first is SDG Indicator 3.8.1, the UHC service coverage index. This index provides a composite measure of coverage of essential services across a set of tracer indicators and serves as the main outcome variable in the report because it offers the clearest internationally comparable measure of access to essential health services over time (United Nations Statistics Division, 2025a; World Health Organization, 2023a).

The second indicator is SDG Indicator 3.8.2, financial protection. This captures the share of the population experiencing catastrophic health expenditure and is included because access to care alone does not provide a complete picture of UHC if households remain exposed to severe financial hardship when using healthcare (United Nations Statistics Division, 2025b; Wagstaff et al., 2018). It therefore functions as a complementary measure of whether service expansion is financially sustainable.

The third indicator is health expenditure per capita. This variable is used as a proxy for health-system investment and is included to explore whether higher levels of spending are associated with stronger service coverage outcomes (World Bank, 2022). In this report, expenditure is used analytically as an associational variable rather than as a direct causal explanation of performance.

The main data sources were the United Nations SDG Global Database, the World Health Organization Global Health Observatory, World Bank classification and development indicator data, and Our World in Data (United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b; World Health Organization, 2023a; World Bank, 2022; Our World in Data, 2026). These were selected because they provide internationally recognised indicators, broad country coverage, and sufficient consistency for long-term comparative analysis.

The analysis period was restricted to 2000–2022. This decision was taken because 2022 represented the most recent year for which the key indicators were relatively comparable across countries and datasets. Although later observations were available for some variables, they were more uneven across countries and indicators due to reporting lags and inconsistent coverage. Restricting the analysis to 2000–2022 therefore improved comparability and reduced the risk of misleading cross-country conclusions based on partial data (World Health Organization and World Bank, 2025).

Before analysis, the datasets were cleaned and standardised. Country names differed across sources, so a country-name harmonisation process was used to match source names with World Bank economy names. World Bank region and income-group classifications were then

attached to the country records to support group-level comparison. The cleaned datasets were then merged into a single country-year analytical dataset containing country, harmonised World Bank country name, year, region, income group, UHC service coverage, financial protection, and health expenditure per capita.

Some missing values remained after the merge because not all indicators are reported annually for all countries. Rather than removing large portions of the data immediately, the merged dataset retained observations wherever possible and used filtering where necessary depending on the analytical task. In grouped regional and income analyses, this also required filtered subsets that excluded unclassified observations where source entities were not present in the World Bank classification file.

The analytical design consists of six main components. First, trend analysis was used to examine how UHC service coverage changed over time globally. This provides the broadest view of whether service coverage improved over the long term.

Second, comparative analysis by region and income group was used to identify structural differences in performance. This directly addresses variation across world regions and income groups and shows whether inequality in progress remains persistent.

Third, country-level summary analysis was used to identify stronger and weaker patterns of progress. This helps distinguish between countries with relatively strong momentum and those where long-term improvement has remained limited. To support this comparison, an average annual rate of change was calculated for each country using the difference between UHC service coverage in 2000 and 2022 divided by the number of years in the interval. This was expressed as:

$$\text{Annual Change 2000–2022} = (\text{UHC 2022} - \text{UHC 2000}) / 22$$

This measure was used to compare long-term momentum in UHC improvement across countries.

Fourth, a cross-sectional analysis of health expenditure and service coverage was used to examine whether higher levels of spending were associated with stronger UHC outcomes. This relationship is interpreted as associative rather than causal because the dataset does not include the broader set of institutional, demographic, and policy variables required to identify full causal drivers of performance (Kruk et al., 2018). It therefore provides a comparative indication of whether stronger UHC coverage tends to be associated with higher health-system spending, without making a causal claim.

Fifth, the report includes a simple forward-looking projection to support consideration of the 2030 objective. Projected UHC values for 2030 were calculated by extending each country's historical average annual rate of change from 2000–2022 forward by eight years. The calculation used was:

$$\text{Projected UHC 2030} = \text{UHC 2022} + (\text{Annual Change 2000–2022} \times 8)$$

This produces an indicative estimate of future trajectory if recent long-term patterns continue. It is intended as an analytical support tool rather than a precise forecast or official SDG attainment model (United Nations, 2015; World Health Organization and World Bank, 2025).

The methodology also includes a descriptive COVID-period perspective. To provide a simple indication of disruption during the pandemic period, recent change was examined using the difference between 2019 and 2021 UHC values:

COVID-period change = UHC 2021 – UHC 2019

This was used as a descriptive proxy for disruption or resilience rather than as a formal estimate of pandemic impact.

Overall, the methodology is designed to balance breadth, comparability, and decision relevance. It provides a structured basis for identifying patterns, comparing performance, and assessing whether observed trajectories appear strong or weak as 2030 approaches

8. Appendices

Appendix A: Data Sources and Indicator Definitions

This report uses three main datasets to assess progress toward Universal Health Coverage (UHC) under SDG 3.8. These datasets were selected because they capture the three core dimensions used throughout the analysis: service coverage, financial protection, and health-system investment. This selection is aligned with the project brief, the project plan, and the methodological direction set out in the kickoff guidance, which emphasised trend analysis, regional and income-group comparison, and the relationship between health spending and UHC progress.

1. SDG Indicator 3.8.1 – UHC Service Coverage Index

This indicator measures coverage of essential health services using a composite index reported on a scale from 0 to 100. The index is based on tracer interventions covering reproductive, maternal, newborn and child health, infectious diseases, non-communicable diseases, and service capacity and access. In this report, SDG 3.8.1 is used as the main outcome indicator because it provides the most direct internationally comparable measure of progress in access to essential health services.

2. SDG Indicator 3.8.2 – Financial Protection

This indicator captures the proportion of the population experiencing large household out-of-pocket health expenditures as a share of total household expenditure or income. It is included because access to care alone does not represent full UHC progress if households remain financially vulnerable when using healthcare. In the global monitoring framework, indicators 3.8.1 and 3.8.2 are intended to be interpreted jointly.

3. Health Expenditure per Capita

This variable represents health spending per person and is used in the report as a proxy for health-system investment. It is included to explore whether higher levels of spending are associated with stronger UHC service coverage outcomes. In the analysis, this measure is treated as an associational explanatory variable rather than as proof of causation.

Main data sources used

- United Nations SDG Global Database
- World Health Organization Global Health Observatory
- World Bank Data / World Development Indicators

- Our World in Data

These are the same core sources identified in the kickoff material and project plan.

Appendix B: Data Preparation and Integration

Before analysis, the datasets were cleaned, standardised, and merged into a single country-year dataset. This stage was necessary because the source datasets used slightly different country naming conventions, varied in time coverage, and required additional classification fields for regional and income-group comparison. The data-preparation process followed the same logic described in the project draft and report methodology.

The main preparation steps were as follows:

1. **Selection of variables**

Only the variables required for analysis were retained from each source dataset. These included country identifiers, year, SDG 3.8.1, SDG 3.8.2, and health expenditure per capita.

2. **Standardisation of country names**

A country-matching process was used to resolve differences in naming conventions across sources. This enabled consistent merging across WHO, UN, and World Bank classification data.

This process was supported through workbook mapping sheets, including country-name harmonisation and identifier standardisation tables.

3. **Assignment of region and income group**

Countries were classified using World Bank economy names, regional groups, and income groups so that comparisons could be made across both geographic and economic dimensions.

4. **Time alignment**

The analysis period was restricted to **2000–2022**. This period was selected because 2022 represented the most recent year for which the key indicators were sufficiently comparable across countries and sources. Later observations were more incomplete and therefore less suitable for consistent cross-country comparison. This was also consistent with the project draft and the methodological rationale used in the report.

5. **Handling of missing values**

Missing observations were retained in the merged dataset where possible, rather than deleting rows immediately. This allowed broad trend analysis to remain intact while enabling more specific filtered analysis for selected tasks such as cross-sectional comparison and correlation.

6. **Creation of merged analytical dataset**

The final merged dataset, referred to as **Master_Merged**, includes:

- Country
- Harmonised World Bank country name
- Year
- Region

- Income group
- UHC service coverage (SDG 3.8.1)
- Financial protection (SDG 3.8.2)
- Health expenditure per capita

A	B	C	D	E	F	G	H	I	J	K	L	M
UN_Code	Country	WB_Name	Year	Region	Income_Group	UHC_3_8_1	Financial_Protection_3_8_2	Health_Expenditure_per_capita	UHC_vis	FP_vis	Spend_vis	Performance_Category
20	Andorra	Andorra	2022	Europe & Central Asia	High income	74.00	4.36	3190.11	74	4.36	3190.11	Emerging Performer
50	Bangladesh	Bangladesh	2022	South Asia	Lower middle income	54.00	41.7	56.62	54	41.7	56.62	Low Performer
64	Bhutan	Bhutan	2022	South Asia	Lower middle income	69.00	8.98	138.01	69	8.98	138.01	Emerging Performer
70	Bosnia and Herzegovina	Bosnia and Herzegovina	2022	Europe & Central Asia	Upper middle income	64.00	8.58	667.3	64	8.58	667.3	Emerging Performer
100	Bulgaria	Bulgaria	2022	Europe & Central Asia	High income	73.00	19.29	1066.69	73	19.29	1066.69	Emerging Performer
120	Cameroon	Cameroon	2022	Sub-Saharan Africa	Lower middle income	47.00	38.84	72.13	47	38.84	72.13	Low Performer
203	Czechia	Czechia	2022	Europe & Central Asia	High income	83.00	11.34	2431.08	83	11.34	2431.08	High Performer
268	Georgia	Georgia	2022	Europe & Central Asia	Upper middle income	71.00	31.75	546.23	71	31.75	546.23	Emerging Performer
300	Greece	Greece	2022	Europe & Central Asia	High income	78.00	14.05	1768.11	78	14.05	1768.11	High Performer
356	India	India	2022	South Asia	Lower middle income	68.00	30.94	81.51	68	30.94	81.51	Emerging Performer
360	Indonesia	Indonesia	2022	East Asia & Pacific	Upper middle income	68.00	27.55	127.44	68	27.55	127.44	Emerging Performer
364	Iran (Islamic Republic of)	Iran, Islamic Rep.	2022	Middle East, North Africa	Upper middle income	81.00	15.65	250.23	81	15.65	250.23	High Performer
392	Japan	Japan	2022	East Asia & Pacific	High income	85.00	10.8	4201.79	85	10.8	4201.79	High Performer
417	Kyrgyzstan	Kyrgyz Republic	2022	Europe & Central Asia	Lower middle income	73.00	9.64	88.08	73	9.64	88.08	Emerging Performer
442	Luxembourg	Luxembourg	2022	Europe & Central Asia	High income	83.00	17.19	7022.63	83	17.19	7022.63	High Performer
458	Malaysia	Malaysia	2022	East Asia & Pacific	Upper middle income	80.00	17.82	458.22	80	17.82	458.22	High Performer
466	Mali	Mali	2022	Sub-Saharan Africa	Low income	41.00	40.07	31.09	41	40.07	31.09	Low Performer
484	Mexico	Mexico	2022	Latin America & Caribbean	Upper middle income	79.00	15.91	650.97	79	15.91	650.97	High Performer
496	Mongolia	Mongolia	2022	East Asia & Pacific	Upper middle income	71.00	27.75	448.24	71	27.75	448.24	Emerging Performer
524	Nepal	Nepal	2022	South Asia	Lower middle income	56.00	24.94	87.34	56	24.94	87.34	Low Performer
566	Nigeria	Nigeria	2022	Sub-Saharan Africa	Lower middle income	47.00	47.35	86.06	47	47.35	86.06	Low Performer
807	North Macedonia	North Macedonia	2022	Europe & Central Asia	Upper middle income	68.00	11.15	561.69	68	11.15	561.69	Emerging Performer
604	Peru	Peru	2022	Latin America & Caribbean	Upper middle income	69.00	28.51	445.1	69	28.51	445.1	Emerging Performer
620	Portugal	Portugal	2022	Europe & Central Asia	High income	84.00	14.58	2600.57	84	14.58	2600.57	High Performer
498	Republic of Moldova	Moldova	2022	Europe & Central Asia	Upper middle income	71.00	10.39	394.42	71	10.39	394.42	Emerging Performer
688	Serbia	Serbia	2022	Europe & Central Asia	Upper middle income	72.00	11.02	843.58	72	11.02	843.58	Emerging Performer

Appendix Figure A1. Example structure of the Master_Merged dataset

This integrated structure provided the analytical basis for trend analysis, regional and income-group comparison, spending-coverage analysis, COVID-period interpretation, and indicative 2030 projection.

Appendix C: Excel Workbook Structure

The analysis was conducted using a structured Excel workbook designed to organise the source data, prepare the merged dataset, and generate analysis outputs for the final report. The workbook combines raw or cleaned source sheets, mapping sheets, merged data, analytical summaries, and report-support visuals.

Core source and preparation sheets

- **UHC_381_clean** – cleaned service coverage dataset for SDG 3.8.1
- **FP_382_Final** – financial protection dataset for SDG 3.8.2
- **Spend_WHO_clean** – health expenditure per capita dataset
- **WB_Class** – World Bank country, region, and income-group classification sheet
- **Country_Name_Fix** – country-name harmonisation table used to align source country names with World Bank economy names
- **Country_Map** – supporting mapping sheet used to standardise country identifiers across datasets
- **Master_Merged** – merged country-year dataset used as the main analysis base

Main analysis sheets

- **Pivot_UHC_Trend** – long-term trend in UHC service coverage
- **Pivot_Region_UHC** – comparison of average UHC service coverage by region
- **Pivot_Income_UHC** – comparison of average UHC service coverage by World Bank income group
- **Scatter_2022** – cross-sectional comparison of health expenditure per capita and UHC service coverage
- **Correlation_Analysis** – summary of the spending-coverage relationship
- **CovidShock_Region** – descriptive comparison of COVID-period UHC change by region
- **CovidShock_IncomeGroup** – descriptive comparison of COVID-period UHC change by income group
- **Country_Progress_Summary** – country-level summary sheet including historical change, average annual change, and projected UHC values to 2030
- **Analysis_Outputs** – supporting summary outputs used for interpretation and cross-checking
- **Table_4_4** – compact summary table used to support the Findings section

The workbook was used as an analytical support tool rather than as an end product in itself. Its main role was to ensure consistent comparison across countries, regions, and income groups while providing a structured basis for the figures and tables reported in the main text.